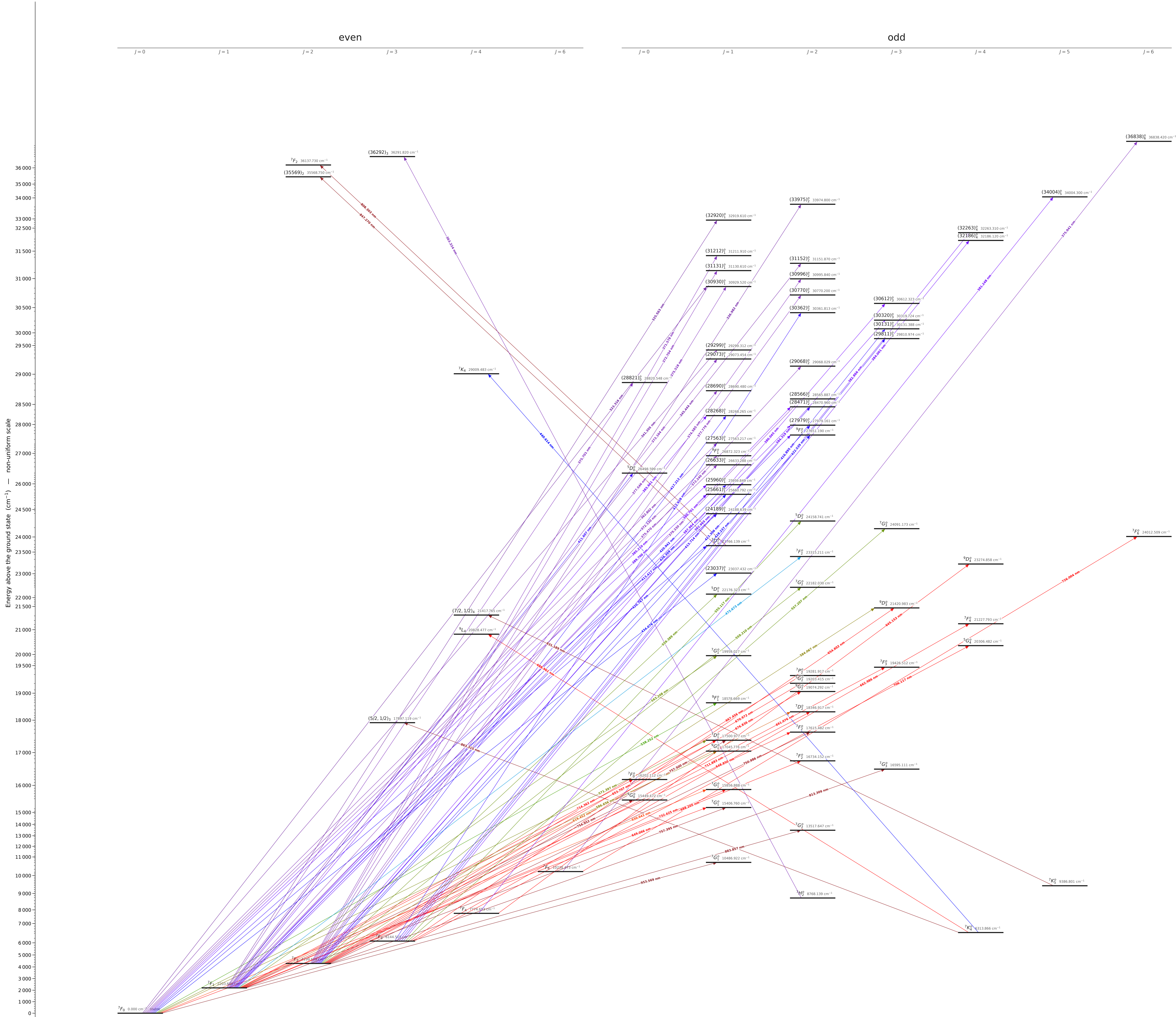
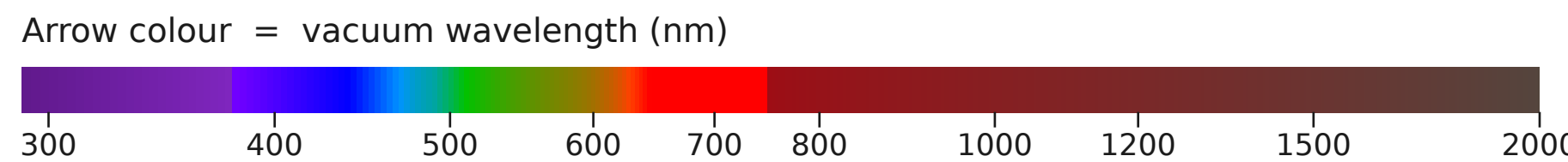


# Plutonium

Pu (Pu I). Classified transitions below 2 μm (levels and lines from the literature): 80 levels, 85 lines.

Pu<sup>+</sup> ionisation limit: 48601.00 cm<sup>-1</sup> = 6.02576 eV (205.757 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $\langle J || e r || J' \rangle$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values: \* = high-accuracy theory, = = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.  
Level caption: energy (from the cited literature: NIST lists no excited levels) and measured lifetime (where one exists).



## Key transitions

| Transition                                                                   | λ vac (nm) | λ air (nm) | ν (THz)  | $\langle U    e r    U' \rangle$ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                                      |
|------------------------------------------------------------------------------|------------|------------|----------|-----------------------------------------------------|----------------------|----------------|------------|--------------------------------------------------------------------------|
| <sup>7</sup> F <sub>0</sub> - ( <sup>2</sup> 5960) <sup>o</sup>              | 385.2103   | 385.1010   | 778.2567 | -                                                   | -                    | -              | -          | 385.210 nm line used for high-resolution in-source resona...             |
| <sup>7</sup> F <sub>1</sub> - ( <sup>2</sup> 7979) <sup>o</sup>              | 387.9645   | 387.8546   | 772.7317 | -                                                   | -                    | -              | -          | 387.965 nm line from the thermally populated 7F <sub>1</sub> level, u... |
| <sup>7</sup> F <sub>0</sub> - ( <sup>2</sup> 4189) <sup>o</sup>              | 413.4172   | 413.3006   | 725.1572 | -                                                   | -                    | -              | -          | first step of the Mainz two-step resonance-ionization sch...             |
| <sup>7</sup> F <sub>0</sub> - ( <sup>2</sup> 3037) <sup>o</sup>              | 420.7667   | 420.6482   | 712.4909 | -                                                   | -                    | -              | -          | first step                                                               |
| <sup>7</sup> F <sub>0</sub> - ( <sup>2</sup> 3037) <sup>o</sup>              | 434.0762   | 433.9542   | 690.6448 | -                                                   | -                    | -              | -          | alternative uranium-free first step                                      |
| <sup>7</sup> D <sub>1</sub> <sup>o</sup> - <sup>7</sup> F <sub>2</sub>       | 808.3035   | 808.0813   | 370.8910 | -                                                   | -                    | -              | -          | second step                                                              |
| <sup>7</sup> D <sub>1</sub> <sup>o</sup> - ( <sup>3</sup> 5569) <sub>1</sub> | 847.2702   | 847.0374   | 353.8334 | -                                                   | -                    | -              | -          | second step                                                              |

λ, ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2τ<sub>u</sub>).

## <sup>241</sup>Pu hyperfine structure (I = 5/2)

| Level                                                           | A (MHz) | B (MHz)  | Shift of each F level from the centre of gravity (MHz)                   |
|-----------------------------------------------------------------|---------|----------|--------------------------------------------------------------------------|
| <sup>7</sup> F <sub>0</sub> - ( <sup>2</sup> 5960) <sup>o</sup> | -2(5)   | 1059(8)  | F=3/2: +748.72 F=5/2: -844.68 F=7/2: +259.15                             |
| <sup>7</sup> F <sub>1</sub> - ( <sup>2</sup> 7979) <sup>o</sup> | -278(6) | 101(18)  | F=3/2: +1043.91 F=5/2: +197.46 F=7/2: -670.05                            |
| <sup>7</sup> F <sub>2</sub> - ( <sup>2</sup> 9797) <sup>o</sup> | 38(3)   | -175(14) | F=1/2: -386.68 F=3/2: -251.20 F=5/2: -69.25 F=7/2: +93.39 F=9/2: +144.65 |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                                                                   | λ vac (nm) | Isotopes | Shift (MHz) |
|------------------------------------------------------------------------------|------------|----------|-------------|
| <sup>7</sup> F <sub>0</sub> - ( <sup>3</sup> 0930) <sup>o</sup>              | 323.316    | 240-239  | -2218.00    |
| <sup>7</sup> F <sub>2</sub> - ( <sup>3</sup> 3975) <sup>o</sup>              | 336.982    | 240-239  | -2398.00    |
| <sup>7</sup> F <sub>0</sub> - ( <sup>2</sup> 9299) <sup>o</sup>              | 341.305    | 240-239  | -1649.00    |
| <sup>7</sup> F <sub>1</sub> - ( <sup>3</sup> 1152) <sup>o</sup>              | 345.444    | 240-239  | -3538.00    |
| <sup>7</sup> F <sub>2</sub> - ( <sup>2</sup> 7563) <sup>o</sup>              | 362.802    | 240-239  | -2218.00    |
| <sup>7</sup> H <sub>2</sub> <sup>o</sup> - ( <sup>3</sup> 6292) <sub>1</sub> | 363.324    | 240-239  | -7525.00    |
| <sup>7</sup> F <sub>2</sub> - ( <sup>3</sup> 1212) <sup>o</sup>              | 371.578    | 240-239  | -2308.00    |
| <sup>7</sup> F <sub>0</sub> - <sup>9</sup> F <sub>1</sub> <sup>o</sup>       | 372.130    | 240-239  | -2069.00    |
| <sup>7</sup> F <sub>1</sub> - ( <sup>2</sup> 9073) <sup>o</sup>              | 372.164    | 240-239  | -1139.00    |
| <sup>7</sup> F <sub>1</sub> - ( <sup>2</sup> 9068) <sup>o</sup>              | 372.240    | 240-239  | -1289.00    |
| <sup>7</sup> F <sub>2</sub> - ( <sup>3</sup> 1131) <sup>o</sup>              | 372.704    | 240-239  | -2188.00    |
| <sup>7</sup> F <sub>2</sub> - ( <sup>3</sup> 0996) <sup>o</sup>              | 374.585    | 240-239  | -2428.00    |
| <sup>7</sup> F <sub>0</sub> - ( <sup>2</sup> 6633) <sup>o</sup>              | 375.470    | 240-239  | -3657.00    |
| <sup>7</sup> F <sub>2</sub> - ( <sup>3</sup> 0930) <sup>o</sup>              | 375.518    | 240-239  | -2368.00    |
| <sup>7</sup> F <sub>0</sub> - ( <sup>3</sup> 6838) <sup>o</sup>              | 375.941    | 240-239  | -3208.00    |
| <sup>7</sup> F <sub>1</sub> - ( <sup>2</sup> 8690) <sup>o</sup>              | 377.546    | 240-239  | -480.00     |
| <sup>7</sup> F <sub>2</sub> - ( <sup>3</sup> 0770) <sup>o</sup>              | 377.779    | 240-239  | -2159.00    |
| <sup>7</sup> F <sub>1</sub> - ( <sup>2</sup> 8566) <sup>o</sup>              | 379.330    | 240-239  | -3418.00    |
| <sup>7</sup> F <sub>2</sub> - ( <sup>3</sup> 0612) <sup>o</sup>              | 380.045    | 240-239  | -2698.00    |
| <sup>7</sup> F <sub>1</sub> - ( <sup>2</sup> 8471) <sup>o</sup>              | 380.701    | 240-239  | -3448.00    |
| <sup>7</sup> F <sub>4</sub> - ( <sup>3</sup> 4004) <sup>o</sup>              | 381.248    | 240-239  | -2338.00    |
| <sup>7</sup> F <sub>2</sub> - ( <sup>3</sup> 2263) <sup>o</sup>              | 382.866    | 240-239  | -1739.00    |

Shift = ν(first isotope) - ν(second isotope).